Inhwa Song • Statement of Purpose

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My research objective lies at the intersection of human-computer interaction (HCI), computational social science (CSS), and natural language processing (NLP). Through my research, I aim to examine the interplay between AI and our society. To this end, I am eager to pursue my Ph.D. at Princeton Computer Science and receive the tutelage of Prof. Manoel Horta Ribeiro.

My research interest has both guided and been shaped by my involvement in many exciting research projects across diverse academic domains and industry settings. While training as a peer counselor, I witnessed firsthand how individuals project and shape their worldviews during personal challenges. Some shared their experiences of using LLM chatbots, highlighting both potential and risks. Motivated by these insights, I seized the opportunity to collaborate with **Prof. Munmun De Choudhury** during my exchange period at Georgia Tech. I led timely research to examine how individuals understand and use LLM chatbots when seeking support for their mental health needs in real-world settings [2]. I conducted semi-structured interviews with 21 global participants by connecting with local forums and support groups. Grounding analysis in psychotherapy literature of what makes support effective, we introduced the concept of therapeutic alignment. Our work has been successfully submitted to *CSCW'24* with me as the first author, currently undergoing review following a request for revisions. This further motivated me to not only envision but also build and test more reliable agents beyond conversational "chat" interfaces.

This motivation led me to a research internship at NAVER AI Lab to work with **Dr. Young-Ho Kim**, whom I have long admired for his novel system contributions to HCI research. I developed ExploreSelf, an LLM-driven interactive system that supports users in exploring their narratives of personal challenges by actively engaging with adaptive guidance [1], grounding in a formative study. Even with my vast hands-on experience managing the development and deployment of computing systems, this particular work was especially valuable to me for three main reasons. First, I gained a deeper understanding of what system contribution means in HCI research. Second, this research I led received a *Revise & Resubmit decision in CHI'25* with compliments for the high research quality. Last but not least, I recognized my ability to quickly learn and apply new technologies and programming methods, along with the confidence that I can effectively tackle any new technical challenges I face in future research.

During my internship at NAVER AI Lab, I could communicate with AI researchers from diverse backgrounds, each bringing a different worldview. For instance, within the goal of responsible AI, stakeholders' perspectives diverged between creating safety benchmarks, embedding safety in LLMs, and investigating what responsible AI means in the context of Korean society. Through direct conversations with stakeholders, I saw firsthand how such divergent views stemmed from the tension between qualitative societal contexts and the quantitative methods that dominate ML. These discussions reinforced my commitment to reveal the dynamics between society and AI. They also motivated me to further tackle research questions that involve direct intervention in language models, beyond translating qualitative insights into concrete system design [2]. Under the guidance of **Prof. Alice Oh**, my ongoing project involves constructing a dialogue dataset of unstructured learner inputs paired with concise queries by fine-tuning the Phi-3 model with the StackExchange dataset.

These research projects gave me the opportunity to adopt a wide range of approaches, from system building and evaluation [1,8], large-scale data analysis [4], fine-tuning language models, to qualitative methods such as semi-structured interviews [2] and cultural probe [3], and also systematic reviews [7]. Above all, I have developed

my ability to identify valuable research questions and learned how to transform an initial idea into systematic research.

Beyond research experience, my community service has been instrumental in shaping my research interests. Through my 2-year experience as the developer of my university's official online community, I witnessed first-hand how subtle design choices in social computing systems can significantly impact the community. Additionally, serving as the student president during the COVID crisis allowed me to hear the voices of underrepresented individuals marginalized due to non-inclusive policies and technology. Together, these experiences have led to my passion for a better understanding of how technology mediates society. Standing on this pivotal societal transition with AI, I aspire to tackle research questions that can practically inform the inclusive policy and design of AI systems.

For example, I hope to examine how generative AI's culturally bound limitations reinforce or shift societal norms. Specifically, I am interested in examining the increasing use of LLMs, often pre-trained on WEIRD[1] data, to study human behavior and how these models influence the representation of diverse cultural perspectives in social science research—an area already overly reliant on WEIRD populations[2]. For instance, in my previous interview study [1], I recruited participants from every inhabited continent, earning recognition from reviewers for securing a culturally diverse sample. This raised an important question for me: If culturally aware language models [7] were used instead of human participants, how would we evaluate that these models adequately represent the culturally diverse samples in the study's context? Building on my prior research, which has broadly explored how technology mediates well-being [1,2,3,4,6], I aim to expand my scope to explore how technology mediates society at scale, with a particular interest in addressing the question above and advancing our understanding of generative AI's societal impact.

With my focus and potential directions to keep in mind, Princeton University would be an ideal place to pursue my doctoral study. I hope to work with **Prof. Manoel Horta Ribeiro** as his research interest, especially in modulating the impact of generative AI in society and using LLM to study human behavior, aligns perfectly with mine. One aspect that drew me to Princeton CS is its collaborative and close-knit community. I also believe that the accessibility of faculty across departments will offer me a rich context for my interdisciplinary research. For instance, I am eager to engage with sociology faculty to deepen the social science dimensions of my projects, and also collaborate with **Prof. Andrés Monroy-Hernández** to incorporate rich insights into the HCI perspective of my work. Upon completing my Ph.D., I aspire to remain in academia as a professor, continuing to research and mentor students to keep investigating the dynamic interplay between emerging technologies and society in the ever-evolving sociotechnical landscape. With my rich research experience, commitment to broadening my worldview, and dedication to fostering societal harmony, I am confident in my ability to succeed in the Princeton CS Ph.D. program and contribute meaningfully to their mission of advancing research for societal impact.

References:

[1] **Inhwa Song**, SoHyun Park, Sachin R. Pendse, Jessica L. Schleider, Munmun De Choudhury, Young-Ho Kim. ExploreSelf: Fostering User-driven Exploration and Reflection on Personal Challenges with Adaptive Guidance by Large Language Models. 1-31. *CHI 2025 (under review with revise & resubmit decision)*. [PDF]

[2] **Inhwa Song***, Sachin R. Pendse*, Neha Kumar, Munmun De Choudhury. The Typing Cure: Experiences with Large Language Model Chatbots for Mental Health Support. 1-21. *CSCW 2024 (under review with major revision decision)*. [PDF]

[1] An acronym for Western, Educated, Industrialized, Rich, and Democratic populations
[2] Henrich, J., Heine, S. J., Norenzayan, A. (2010). Most people are not WEIRD. Nature, 466(7302), 29-29.

[3] **Inhwa Song**, Kwangyoung Lee, Janghee Cho, Amon Rapp, Hwajung Hong. Understanding God-Saeng Phenomenon: Young Generation's Pursuit of Authentic Well-being. 1-17. *(in prep for submission to DIS 2025, submitted but not accepted to DIS 2024)*.

[4] **Inhwa Song***, David Gomez*, Ray Hung*, Charles Nimo*, (2023) Deep Learning for Identifying Language Features that Differentiate Mental Health Communities on Reddit. 1-8. *unpublished*. [PDF]

[5] Janghee Cho, **Inhwa Song**, Zainab Agha, Bengisu Cagiltay, Veena Calambur, Minjin "MJ" Rheu, Jina Huh-Yoo, Mobile Technology and Teens: Understanding the Changing Needs of Sociocultural and Technical Landscape. 1-11. *CHI 2025 Workshop Proposal (accepted)*. [PDF]

[6] **Inhwa Song**, Kwangyoung Lee, Janghee Cho, Amon Rapp, Hwajung Hong (2024) Bridging Values and Technology in God-Saeng Phenomenon: Designing AI for Self-Directed Well-being. 1-3. *CHI 2024 Workshop: Designing (with) AI For Wellbeing (accepted)*. [PDF]

[7] Siddhesh Pawar*, Junyeong Park*, Jiho Jin, Arnav Arora, Junho Myung, Srishti Yadav, **Inhwa Song**, Alice Oh, Isabelle Augenstein (2024) Survey of Cultural Awareness in Language Models: Text and Beyond. 1-87. *Journal of Computational Linguistics (under review)*. [PDF]

[8] Haesoo Kim, **Inhwa Song**, Juho Kim. (2022) Learnersourcing Modular and Dynamic Multiple Choice Questions. 1-5. *L@S 2022 Workshop on Learnersourcing Student Generated Content (accepted)*. [PDF]

Inhwa Song: Curriculum Vitae

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https://inhwasong.com
https://github.com/greenina
Google Scholar (Link)

Applying to Princeton University, Computer Science, Ph.D.

Undergraduate & Researcher @ KAIST

EDUCATION

KAIST (Korean Advanced Institute of Science and Technology) — **Daejeon, Republic of Korea**
B.S. Computer Science, Advisor: Alice Oh — Feb 2019 – June 2025 (Expected)
Relevant courses include: Social Computing(A0), Social Network Analysis(A+), Intro to Deep Learning(A+), Deep Reinforcement Learning (A0), Intro to HCI(A-), Data Visualization(A0), Operating Systems and Lab(A0)

Georgia Institute of Technology — **Atlanta, Georgia**
Exchange Student, Computer Science, Research Advisor: Munmun De Choudhury — Aug 2023 – Dec 2023
Courses: Deep Learning(A), Knowledge-based AI(A), Computer Vision(A), Undergraduate Research(S)

RESEARCH EXPERIENCE

Users & Information Lab, KAIST [p.5] — **Daejeon, Republic of Korea**
Undergraduate Researcher, *Advisor: Prof. Alice Oh* — Sep 2024- present
Role: Fine-tuned a Phi-3 language model with pre-processed StackExchange post data to create a multi-turn dialogue dataset, enabling the transformation of unstructured learner inputs into concise queries and providing keywords for guided exploration in educational contexts (work-in-progress).
Methods and Tools: Fine-tuning open-sourced language model, Data collection and preprocessing.

Role: Co-authored a survey paper on cultural awareness in language models, managing sections on 1) culturally aware machine translation and 2) culturally aware dialogue.
Methods and Tools: Systematic reviews, Publication drafting (in collaboration).

AI Lab, NAVER [p.7] — **Seong-Nam, Republic of Korea**
Research Intern, *Advisor: Dr. Young-Ho Kim* — Mar 2024- Sep 2024
Role: Led the technical implementation and design of an LLM-driven interactive system that supports user-driven exploration and reflection of personal challenges, drawing on expert interviews from mental health professionals.
Methods and Tools: End-to-end system building, User testing, Interaction log analysis (quantitative & qualitative), Semi-structured interview, IRB Approval, Publication drafting and submission, Addressing peer reviews.

Social Dynamics and Wellbeing Lab (SocWeB Lab), Georgia Tech [p.6] — **Atlanta, Georgia**
Visiting Undergraduate Researcher, *Advisor: Prof. Munmun De Choudhury* — Aug 2023- Jan 2024
Role: Built on interviews with 21 individuals from globally diverse backgrounds to analyze how users create unique support roles for LLM chatbots, fill in gaps in everyday care, and navigate associated cultural limitations when seeking support from chatbots, grounding analysis in psychotherapy literature.
Methods and Tools: Global recruitment, Semi-structured Interview, IRB Approval, Thematic analysis, Publication drafting and submission, Addressing peer reviews.

Data, Interaction, and Design Lab (DxD Lab), KAIST [p.2][p.4] — **Daejeon, Republic of Korea**
Undergraduate Research Intern, *Advisors: Prof. Hwajung Hong, Prof. Janghee Cho* — Oct 2022- Aug 2023
Role: Led the design and conduction of a 10-day cultural probe exploring young people's perceptions of well-being phenomena in South Korea, focusing on the (mis)alignment of personal informatics technology use with their existential matters.
Methods and Tools: Cultural probe design, Semi-structured interview, IRB Approval, Thematic analysis, Publication drafting and submission, Addressing peer reviews.

KAIST Interaction Lab (KIXLab), KAIST [p.1] **Daejeon, Republic of Korea**

Undergraduate Research Intern, *Advisor: Prof. Juho Kim* June 2021- Sep 2022

Role: Led the real-class deployment of an online education crowdsourcing system for modular and dynamic generation of multiple-choice questions. More than 30 students synchronously accessed the system during the in-class activity.
Methods and Tools: Real-class deployment, End-to-end system building, Iterative prototyping, Publication drafting and submission (in collaboration).

PUBLICATIONS

[p.7] ExploreSelf: Fostering User-driven Exploration and Reflection on Personal Challenges with Adaptive Guidance by Large Language Models

Inhwa Song, SoHyun Park, Sachin R. Pendse, Jessica L. Schleider, Munmun De Choudhury, Young-Ho Kim
CHI'25 (under review, R&R decision). 1-31. [PDF]

[p.6] The Typing Cure: Experiences with Large Language Model Chatbots for Mental Health Support

Inhwa Song*, Sachin R. Pendse*, Neha Kumar, Munmun De Choudhury
CSCW'24 (under review, major revision decision). 1-21. [PDF]

[p.5] Survey of Cultural Awareness in Language Models: Text and Beyond

Siddhesh Pawar*, Junyeong Park*, Jiho Jin, Arnav Arora, Junho Myung, Srishti Yadav, **Inhwa Song**, Alice Oh, Isabelle Augenstein
Journal of Computational Linguistics (under review). 1-87. [PDF]

[p.4] Understanding God-Saeng (God生) Phenomenon: Young Generation's Pursuit of Authentic Well-being,

Inhwa Song, Kwangyoung Lee, Janghee Cho, Amon Rapp, Hwajung Hong
In prep for DIS'25, submitted and rejected in DIS'24. 1-17. [PDF]

[p.3] Mobile Technology and Teens: Understanding the Changing Needs of Sociocultural and Technical Landscape

Janghee Cho, **Inhwa Song**, Zainab Agha, Bengisu Cagiltay, Veena Calambur, Minjin "MJ" Rheu, Jina Huh-Yoo
CHI'25 Workshop Proposal (conditionally accepted). 1-12. [PDF]

[p.2] Bridging Values and Technology in God-Saeng Phenomenon: Designing AI for Self-directed Well-being

Inhwa Song, Kwangyoung Lee, Janghee Cho, Amon Rapp, Hwajung Hong
CHI'24 Workshop on Designing (with) AI for Wellbeing (accepted). 1-3. [PDF]

[p.1] Learnersourcing Modular and Dynamic Multiple Choice Questions

Haesoo Kim, **Inhwa Song**, Juho Kim
L@S'22 Workshop on Learnersourcing Student Generated Content (accepted). 1-5. [PDF]

FELLOWSHIPS & AWARDS

KAIST Undergraduate Research Grant, 2M KRW 2023
Conference Scholarships from KAIST (CHI `23 `24, NeurIPS `23, CSCW'23, UIST `23, FAccT `22) 2022 - 2024
Lim Mi-Sook Scholarship, 1M KRW, Honored by Prof. Kyunghyun Cho @NYU 2021
Superior Leadership Award, KAIST, Top 3% 2020
Grand Prize, World Friends Tanzania Outstanding ICT Volunteer (in a team of 16) 2020
Best Freshmen Award, KAIST 2019
Merit-based scholarship for undergraduate studies funded by the Korean government (full tuition for four years) 2019 - 2023
Hanseong Nobel Scholarship for Gifted in STEM, 5M KRW 2018

PROFESSIONAL SERVICES & LEADERSHIP

Reviewer, ACM CHI'25 2024
Student President of KAIST School of Computing, over 900 students Feb 2021 - Feb 2022
Student Volunteer, ACM CSCW `23 (in-person) Oct 2023
Software Engineer, System Programmers' Association for Researching Computer Systems June 2020 - Aug 2023

- Developed the frontend/backend of the KAIST official online community website (ARA) (in collaboration), attracting over 400 visitors per day (https://ara.kaist.ac.kr)

Local Committee Support, ACM FAccT `22 (in-person) 2022

Student Organizer, Rising Stars in AI/CS/EE Workshop (Sponsored by Google) 2022
Vice President of Public Relations, Daedeok Toastmasters Club of Public Speaking Feb 2023 - Nov 2023

TALKS & PRESS

Pre-ACL Workshop, Experiences with Large Language Model Chatbots for Mental Health Support June 2024
KAIST Human of CS [link (English)] May 2023
KAIST Times: My Quantified Self Story [link (Korean)] Mar 2023
Authoring Algorithm for Personal Daily Score (hosted by Open Humans & QuantifiedSelf) [wiki, video (English)] Oct 2022
KAIST 100yr Vision Magazine Aug 2021
KAIST Vision Quarterly [link (Korean)] Mar 2021
Daehak Journal Webzine [link (Korean)] Oct 2019

WORK EXPERIENCE

Chief Technical Officer, **HELPO** April 2020 - Dec 2020

- Led the development of an online platform for eliciting knowledge graphs from video sequences to support interactive learning in video learning platforms.

TEACHING /MENTORING

Teaching Assistant KAIST CS496 Immerse Coding Camp Dec 2020 - Jan 2021

Mentoring

- Mentored pediatric cancer children—Taught basic physics & biology Mar 2021 - Sep 2021
- Tutored mathematics for high school students Mar 2019 - Aug 2020

SKILLS

Below are selected skills. For detailed project descriptions, visit inhwasong.com.

Programming Languages & Library

- Advanced: Typescript, Python, MongoDB, Pytorch, Hugging Face (transformers, datasets, tokenizers), ReactJS, NodeJS, Redux, Plotly
- Moderate: C, Kotlin, MySQL, VueJS, TensorFlow
- Novice: Scala, C#, Flutter

Research Methods/Skills

- Technical: End-to-end System Building, Large-scale Data Analysis, Building/Fine-tuning Deep Learning Models
- Qualitative: Thematic Analysis, Semi-structured Interview, Cultural Probe

Languages

- Korean (Native), English (Fluent)